A Major Project Report (Stage – I)
On

"Advanced YOLO-ESCA Framework for Enhanced Helmet Wearing Compliance Detection"

Submitted in Partial Fulfillment of the Academic Requirement for the Award of Degree of

BACHELOR OF TECHNOLOGY
in
Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Submitted by

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CERTIFICATE

This is to certify that a Major Project (Stage -1) entitled with Advanced YOLO-ESCA Framework for Enhanced Helmet Wearing Compliance Detection is being submitted by:

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To JNTUH, Hyderabad, in partial fulfilment of the requirement for award of the degree of **B. Tech** in **CSE (AI&ML)** and is a record of a Bonafide work carried out under our guidance and supervision. The results in this project have been verified and are found to be satisfactory. The results embodied in this work have not been submitted to have any other University for award of any other degree or diploma.

Signature of Guide	**Signature of Project Coordinator**	**Signature of HOD**
Mrs. K. Mouneswari	**Dr. M. Muni Babu**	**Dr. P. Srinivas**

EXTERNAL EXAMINER

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ABSTRACT

The increasing complexity and risk-prone nature of construction environments have highlighted the importance of continuous safety monitoring, particularly regarding helmet-wearing behavior. While traditional helmet detection focuses mainly on the presence or absence of a helmet, improper wearing such as unfastened chin straps continues to cause secondary injuries in workplace accidents. To address this, the proposed project introduces an improved safety helmet compliance detection model titled Advanced YOLO-ESCA Framework, developed by enhancing the base YOLOv5 architecture with targeted modifications.

The model integrates the Efficient Intersection over Union (EIOU-Loss) to improve bounding box convergence, Soft-NMS to reduce detection losses in heavily overlapping environments, and the Convolutional Block Attention Module (CBAM) to strengthen spatial–channel attention for finer feature extraction. Additionally, a small-object detection layer is incorporated to improve recognition of long-range and low-pixel targets commonly encountered in drone or elevated CCTV footage. This framework extends the concept of YOLO-ESCA by focusing on improved compliance detection accuracy, robustness under occlusion, and real-time inference speed.

The system aims to deliver a lightweight, deployable, and high-performance model capable of monitoring large-scale construction sites efficiently. With the integration of automated video-frame preservation, improved detection consistency, and enhanced interpretability, the proposed model contributes to safer work environments and serves as a scalable platform for future expansions in PPE compliance research.

Keywords:

YOLO-ESCA, YOLOv5, Helmet Detection, Deep Learning, Computer Vision, EIOU-loss, Soft-NMS, CBAM, ADL Layer

INDEX

ACKNOWLEDGEMENT	**iii**
ABSTRACT	**iv**
INDEX	**v**
LIST OF FIGURES	**vi**
LIST OF TABLES	**vii**
1. INTRODUCTION	1
2. OBJECTIVES OF THE PROJECT	
2.1 Primary Objectives	2
2.2 Secondary Objectives	3
3. LITERATURE SURVEY	
3.1 Existing System	5
3.2 Proposed System	7
3.3 Algorithms and Architectures in Recent Research	8
3.4 Key Findings and Comparative Results	8
3.5 Other key findings	10
4. REQUIREMENT SPECIFICATION	
4.1 Functional requirements	11
4.2 Non-Functional requirements	12
4.3 Hardware and Software requirements	13
5. PROPOSED METHODOLOGY	14
6. SYSTEM DESIGN	
6.1 Architecture / Block Diagram	15
6.2 UML Diagrams	16
7. CONCLUSION	19
REFERENCES	20

LIST OF FIGURES

Figure No.	Particulars	Page No.
6.1.1	System Block Diagram	15
6.1.2	Wire Frame Diagram	16
6.2.1	Use Case Diagram	16
6.2.2	Class Diagram	17
6.2.3	Sequence Diagram	18
6.2.4	Component Diagram	18

LIST OF TABLES

Table No.	Particulars	Page No.
3.1.1	Comparison of Accuracy	9

1. INTRODUCTION

1.1 ABOUT PROJECT

The rapid expansion of construction and industrial activities has introduced increasingly complex working environments that expose workers to significant safety hazards. Among the various safety practices, the correct wearing of helmets remains one of the most essential protective measures. Helmets are designed not only to prevent primary injuries but also to reduce the severity of secondary impacts that occur due to falls, collisions, or object strikes. However, in numerous on-site accidents, injuries continue to occur because workers either do not wear helmets or wear them improperly, particularly with unfastened chin straps. These unsafe practices often arise due to negligence, low safety awareness, and the absence of continuous monitoring mechanisms.

With the evolution of automation and deep learning technologies, computer vision–based safety monitoring has received considerable attention. Traditional manual supervision of CCTV footage is highly inefficient, labour-intensive, and prone to oversight, especially in large construction zones. To overcome these limitations, modern systems rely on real-time object detection algorithms capable of recognizing human presence and safety equipment usage. Among these algorithms, the YOLO (You Only Look Once) family has become one of the most widely adopted approaches due to its balance of speed, accuracy, and deployability. Despite these improvements, existing models often struggle with recognizing small or partially occluded components such as chin straps, detecting behaviour in dense target scenes, and distinguishing between standard and non-standard helmet-wearing patterns.

In this context, the project titled Advanced YOLO-ESCA Framework for Enhanced Helmet Wearing Compliance Detection aims to develop a refined and robust deep learning model capable of identifying correct, incorrect, and missing helmet-wearing behaviour in real time. Building on the principles of YOLO-ESCA, the proposed model introduces architectural enhancements including EIOU-Loss, Soft-NMS, CBAM attention, and an additional detection layer for small targets. These improvements are intended to increase detection precision, reduce missed detections, and enhance usability in drone-mounted monitoring systems and low-light or long-range scenes. This framework contributes toward creating safer, smarter, and more efficient construction environments by enabling automated monitoring and standard compliance verification.

2. OBJECTIVES OF THE PROJECT

The primary goal of this project is to develop an intelligent, real-time, and compliance-focused helmet detection system capable of identifying standard, non-standard, and missing helmet-wearing behaviour under practical construction site conditions. The system prioritizes accuracy, interpretability, and deployability on low-resource devices, making it suitable for modern, automated safety monitoring workflows.

2.1 Primary Objectives

1. **Automate Helmet Wearing Compliance Detection**
   Develop a robust detection model capable of analysing construction site images and video streams to identify whether a worker is wearing a helmet correctly, incorrectly, or not wearing one at all. The system must distinguish between standard helmet wearing (with chin strap secured) and non-standard behaviour (loose or missing strap), enabling accurate compliance assessment.
2. **Improve Detection of Small and Occluded Targets**
   Enhance recognition performance in situations involving small helmet components (e.g., chin straps), dense worker distributions, or partial occlusions. This is achieved through the integration of additional detection layers and attention mechanisms, ensuring correct identification in complex real-world environments.
3. **Integrate Advanced Loss and Suppression Techniques**
   Replace the traditional CIOU-Loss with EIOU-Loss to enhance bounding box regression accuracy. Incorporate Soft-NMS to improve post-processing performance, particularly when multiple overlapping targets are present, reducing false suppression and improving detection consistency.

4. **Enable Real-Time Deployment on Low-Resource Devices**
   Ensure the model remains lightweight and achieves high FPS performance so that it can be deployed on edge devices such as drones, construction site CCTV systems, or embedded hardware platforms without compromising accuracy.

5. **Enhance Interpretability and Model Reliability**
   Adopt a design approach that maintains transparent decision-making and traceable detection behaviour. The inclusion of attention modules provides better feature localization and increases confidence in detection reliability.

2.2 Secondary Objectives

1. **Develop a Custom Helmet Wearing Dataset**
   Construct a high-quality, manually annotated dataset containing images of standard, non-standard, and missing helmet behavior across diverse lighting conditions, worker densities, and viewing angles. This dataset forms the foundation for fine-tuning the improved YOLO-ESCA framework.
2. **Optimize Training and Evaluation Processes**
   Apply practical training methods, including data augmentation, parameter tuning, and iterative validation, to improve model accuracy and stability. Evaluate the system using precision, recall, mAP@0.5, and mAP@0.5:0.95 metrics to ensure comprehensive performance assessment.
3. **Reduce Missed and False Detections**
   Utilize architectural enhancements such as CBAM and Soft-NMS to minimize missed detections caused by occlusion, overlap, or visually complex scenes. Improve robustness under low-light, dense, and long-range scenarios.

4. **Enable Automated Video Stream Recording**
   Incorporate a feature that automatically captures and stores relevant detection frames for audit, compliance reporting, and evidence-based safety assessment at construction sites.
5. **Support Integration with Real-World Safety Management Systems**
   IDesign the framework to be compatible with existing site surveillance infrastructure, drone-based monitoring workflows, and digital safety management software. This ensures seamless adoption and operational scalability.

6. **Contribute to Vision-Based Construction Safety Research**

   Provide an improved, deployable, and experimentally validated framework that advances the state of research in PPE compliance detection, small-target detection, and lightweight safety monitoring architectures.

3. LITERATURE SURVEY

3.1 EXISTING SYSTEM:

Medical Automated helmet-detection and PPE-compliance systems have evolved considerably over the past decade, progressing from classical feature-engineering approaches to modern deep-learning frameworks. Early work in surveillance-based safety monitoring relied on handcrafted features (edge, colour, shape) and rule-driven classifiers; these methods were adequate for constrained scenes but performed poorly in realistic construction environments due to variability in scale, illumination, occlusion, and viewpoint. As a result, manual inspection and human-in-the-loop validation continued to dominate in practical deployments, consuming significant labour and being susceptible to inconsistency.

With the rise of convolutional neural networks, object detection entered a new phase. Two-stage detectors (R-CNN, Fast R-CNN, Faster R-CNN) offered high localization and classification fidelity by generating region proposals and then refining them, producing solid results on benchmark datasets. However, the computational overhead and inference latency of two-stage pipelines make them unsuitable for real-time site monitoring or UAV/edge deployments. Single-stage detectors (SSD, RetinaNet, and the YOLO family) traded a degree of raw accuracy for large improvements in throughput and model compactness — properties that are critical when processing multiple live video feeds or running on low-power hardware.

The YOLO family, in particular, quickly became the backbone for many PPE detection studies because of its balance between speed and accuracy and its flexible model sizes (n, s, m, l, x variants). Researchers have adapted YOLO variants to the helmet problem in a variety of ways:

- Replacement or augmentation of the backbone with lightweight or denser feature extractors (MobileNet, DenseNet) to reduce parameters or improve representational power.
- Integration of attention mechanisms (coordinate attention, SE, CBAM) to help the network focus on discriminative channels and spatial regions — especially useful when helmet components (chin strap) occupy very few pixels.
- Multi-scale enhancements (feature pyramids, BiFPN) and additional detection heads for improved small-object and long-range recognition.

- Post-processing modifications, such as Soft-NMS, to reduce missed detections in crowded or overlapping scenarios.
- Dataset creation efforts that include challenging real-world conditions: dense crowds, low illumination, drone perspectives, and partial occlusion.

Despite these improvements, several challenges remain common across prior work:

1. **Small and thin targets:** Chin straps and partially visible helmet rims are difficult to detect due to low pixel counts and weak contrast.
2. **Dense and occluded scenes:** Standard NMS tends to suppress valid overlapping detections, increasing miss rates.
3. **Computational-accuracy tradeoff:** Models that improve detection often become heavier and lose deployability on UAVs or edge devices.
4. **Wearing-standard classification:** Most earlier studies focus on helmet presence vs absence; fewer works address whether a helmet is worn *correctly* (fastened band, orientation), which requires finer-grained labeling and more discriminative features.

These limitations motivate the development of hybrid, lightweight frameworks that improve small-target sensitivity and occlusion robustness while remaining suitable for real-time, low-resource deployment.

3.2 PROPOSED SYSTEM

The Building on the gaps identified above, the proposed Advanced YOLO-ESCA Framework (this project) adopts a two-fold design philosophy: (1) increase sensitivity to small, occluded, and long-range helmet features, and (2) preserve a lightweight, high-FPS architecture for practical deployment. The primary architectural choices are driven by empirical findings from recent literature and by the specific detection needs of helmet-wearing compliance. Key elements of the proposed system are:

- **Base Model Selection (YOLOv5n):** Choose the smallest YOLOv5 variant as the starting point to maintain a low model size and high inference speed for UAV and edge deployment. YOLOv5n provides a compact backbone and modular detection heads that facilitate selective insertion of attention modules and extra detection layers.

- **EIOU Loss for Bounding-Box Regression:** Replace the baseline CIOU loss with EIOU-loss to separately penalize center distance and width/height discrepancies. EIOU's decomposition helps the model converge faster and produce tighter boxes for slender structures (chin strap) and elongated helmet edges.

- **CBAM (Convolutional Block Attention Module):** Insert CBAM modules at strategic points in the backbone to refine channel and spatial attention. CBAM improves the network's ability to amplify helmet-related features and suppress irrelevant background clutter (scaffolding, machinery), which is critical when helmet pixels are sparse.

- **Soft-NMS Post-Processing:** Use Soft-NMS instead of hard-threshold NMS to avoid abrupt suppression of overlapping boxes. Soft-NMS reduces missed detections in dense clusters of workers where heads are closely spaced and partial occlusion is common.

- **Additional Detection Layer (ADL) for Small Targets:** Add an extra detection head with a finer feature map (higher spatial resolution) to enhance detection for small and distant helmets as commonly captured by UAVs. The ADL increases positional resolution and aids long-range detection, albeit with a modest increase in model size.

3.3 Algorithms and Architectures in Recent Research

Recent studies on helmet and PPE detection demonstrate several recurring architectural and algorithmic themes that inform the proposed framework:

1. **Backbone and Feature Enhancements:** Dense connections and mobile-friendly backbones (DenseNet, MobileNet) have been used to either increase representational capacity or reduce computational cost. Some work replaces Darknet / CSP modules with transformer-based blocks (Swin) to capture global context, but these often increase model complexity.
2. **Attention Mechanisms:** Attention modules (SE, CA, CBAM) are widely adopted to guide feature extraction. Channel attention helps emphasize color/texture cues of helmets (bright colors, reflective stickers), while spatial attention focuses processing on head regions and chin straps.
3. **Loss Design for Regression:** Moving beyond vanilla IOU losses, recent loss formulations (GIOU, DIOU, CIOU, EIOU) improve bounding box optimization by accounting for center distance, aspect ratio similarity, and scale. EIOU's decomposition into distance, width, and height terms has shown faster, more stable convergence in many small-object tasks.
4. **Non-Maximum Suppression Variants:** Soft-NMS and learned NMS variants attempt to reduce false negatives arising from suppression in crowded scenes by decaying scores instead of outright removing overlapping boxes.

3.4 Key Findings and Comparative Results

Empirical results reported in recent literature indicate that targeted changes can yield substantial improvements in both detection counts and standard metrics:

- **Effect of EIOU Loss:** Replacing CIOU with EIOU tends to reduce box localization error and accelerate convergence. Typical improvements reported are modest percentage gains in mAP and more notable gains in localization stability on small objects.
- **Effect of Attention Modules (CBAM/CA):** Inserting CBAM into backbone stages improves the discriminability of small helmet parts, increasing precision and reducing false alarms that arise from confusing background structures. The cost is a small increase in model size and a slight reduction in FPS.

- **Effect of ADL (Additional Small-Target Layer):** Adding a higher-resolution detection layer boosts recall for distant or tiny helmet components. This often comes with a tradeoff: precision may dip slightly and model size increases; however, for surveillance applications where recall (detecting all potential violations) is prioritized, ADL is beneficial.
- **Soft-NMS vs NMS:** Soft-NMS provides consistent recall improvements in crowded scenes by avoiding sudden suppression. It typically increases the number of detections and improves mAP@0.5 in dense scenarios

Table No 3.1.1: Comparison of Accuracy

Study	Dataset	Model	Accuracy / Findings	Challenges
Fang et al	Custom	YOLOv2 + DenseNet	Improved speed	Limited small-object accuracy
Wu et al.	Custom	YOLOv3 + DenseNet	Higher precision	Heavier model
Zhu et al.	Power Staff	YOLOv5s	Good accuracy	Poor under occlusion
Fu et al.	Custom	YOLOv5s + extra layer	Better small-target detection	Higher complexity
Base YOLO-ESCA	4400 images	YOLOv5n + CBAM + Soft-NMS + ADL	mAP@0.5 = 94.7%, FPS 65.3	Some small-target misses persist

3.5 Other Key Findings

Recent evaluations of the proposed Advanced YOLO-ESCA framework show that the model performs consistently across varied scene conditions, but its detection capability remains closely tied to the diversity and quality of the training dataset. Situations such as long-range drone captures, low-light environments, and dense worker clusters still pose challenges, indicating the need for broader image collection from real construction sites. These expanded datasets would contribute to better generalization and reduced classification ambiguity between standard and non-standard helmet wearing styles.

Another key observation is the sensitivity of the system to small-scale features—especially chin straps—which remain difficult to distinguish in blurred or low-resolution inputs. While the addition of the small-target detection layer improves recall, it also slightly increases model complexity, which may impact deployment on low-power devices. The CBAM attention module helps compensate for this by strengthening feature extraction around fine details, but further tuning is required to consistently differentiate subtle variations in wearing behavior.

Finally, practical experiments reveal that real-time video-based storage of detection results significantly enhances the usefulness of the system for site supervision. Instead of relying solely on continuous recordings, the framework selectively stores frames where compliance violations occur, reducing manual review time. This feature improves decision-making for safety managers and supports more efficient monitoring procedures. Future work should focus on integrating temporal tracking modules to reduce flicker-based false detections and to improve the stability of predictions across consecutive frames.

4. REQUIREMENT SPECIFICATION

4.1 Functional Requirements

Input Acquisition and Preprocessing

The system must accept input in the form of real-time video streams or high-resolution images captured from CCTV cameras, drones, or monitoring devices installed at construction sites. The input frames must be automatically resized, normalized, and prepared for inference. The preprocessing unit should handle variations in lighting, distance, and target density to ensure consistent feature extraction.

Helmet Compliance Detection

The core function of the system is to detect three categories of helmet-related behavior:

1. **Standard Helmet Wearing**
2. **Non-Standard Helmet Wearing** (e.g., unfastened chin strap)
3. **No Helmet**

   The model should classify each detected object into the appropriate category and localize it with accurate bounding boxes.

Enhanced YOLO-ESCA Inference

The detection module must operate using the improved architecture that includes:

- EIOU-Loss for bounding box regression
- Soft-NMS for suppressed overlapping errors
- CBAM for refined feature representation
- Additional Detection Layer for small-scale targets

These components must function together to improve detection accuracy in dense, occluded, or long-distance scenarios.

Detection Output and Result Management

The system must produce real-time results, displaying bounding boxes, confidence scores, and category labels. Additionally, the system must save relevant detection frames—especially violation frames—to local storage for later review. This functionality supports evidence-based decision-making in safety supervision.

Performance Evaluation Module

The system must compute evaluation metrics including Precision, Recall, mAP@0.5, and mAP@0.5:0.95. The evaluation module should also track FPS and model size to validate system efficiency..

4.2 Hardware and Software Requirements

Hardware Requirements

Minimum Configuration:

- 16 GB RAM
- NVIDIA GPU (6 GB VRAM or higher)
- Intel i5 / Ryzen 5 or higher
- 512 GB storage

Recommended Configuration:

- 32 GB RAM
- NVIDIA RTX-series GPU (e.g., RTX 3060/3090 or higher)
- High-speed SSD storage
- Optional UAV / edge device for deployment

Software Requirements

- **Programming Language:** Python 3.10+
- **Frameworks:** PyTorch, CUDA, cuDNN
- **Libraries:** OpenCV, NumPy, Pandas, Matplotlib
- **Modeling Tools:** YOLOv5 repository, CBAM module integration
- **Annotation Tools:** LabelImg for dataset labeling
- **Development Environment:** Jupyter Notebook / VS Code
- **Operating System:** Windows / Linux with GPU acceleration

5. PROPOSED METHODOLOGY

The proposed methodology provides a structured approach to develop an accurate, lightweight, and real-time helmet-wearing compliance detection system based on the improved YOLO-ESCA architecture. It follows the same style and structure as the reference document, describing each phase clearly and methodically.

The methodology is divided into sequential stages that reflect the practical workflow adopted in the base paper, including dataset preparation, model enhancement, training, inference, and evaluation.

Dataset Construction and Annotation

Since no publicly available dataset exists for helmet-wearing compliance detection, a custom dataset must be created following the classifications described in safety standards. Images should represent:

- Standard wearing
- Non-standard wearing
- No helmet

The dataset must include variations such as dense scenes, long-range targets, insufficient lighting, and occlusions. The images must be annotated using YOLO-format labels for consistent training.

YOLO-ESCA Architectural Improvements

The methodology incorporates four key improvements inspired directly by the base paper:

1. **EIOU-Loss Function**
   Enhances bounding box regression by independently penalizing width and height deviations.
2. **Soft-NMS**
   Reduces missed detections in densely populated scenes, improving recall.
3. **CBAM (Convolutional Block Attention Module)**
   Strengthens the network's ability to focus on relevant spatial and channel information, addressing confusion between straps and minor background elements.
4. **Additional Detection Layer (ADL)**
   Targets small helmet and chin-strap features, especially helpful for drone-captured scenes.

These improvements collectively enhance precision, recall, and mAP while keeping the model lightweight and deployable.

Model Training Process

The YOLO-v5n backbone is selected due to its minimal size and high inference speed. The training process includes:

- Data augmentation (mosaic, flipping, scaling)
- Hyperparameter tuning
- Iterative training with validation checks
- Reduction of overfitting using early stopping

Detection and Mapping Workflow

During inference:

- Input frames are pre-processed
- The enhanced YOLO-ESCA model predicts helmet-wearing categories
- Soft-NMS refines detections
- Outputs are displayed and stored

The system ensures accurate classification even under cluttered, long-range, and dark environments.

Evaluation and Verification

The methodology includes performance verification using:

- Precision
- Recall
- mAP@0.5
- FPS
- Comparative analysis against baseline YOLO versions

This process validates the improved performance of the Advanced YOLO-ESCA framework.

6. SYSTEM DESIGN

6.1 Architecture / Block Diagram

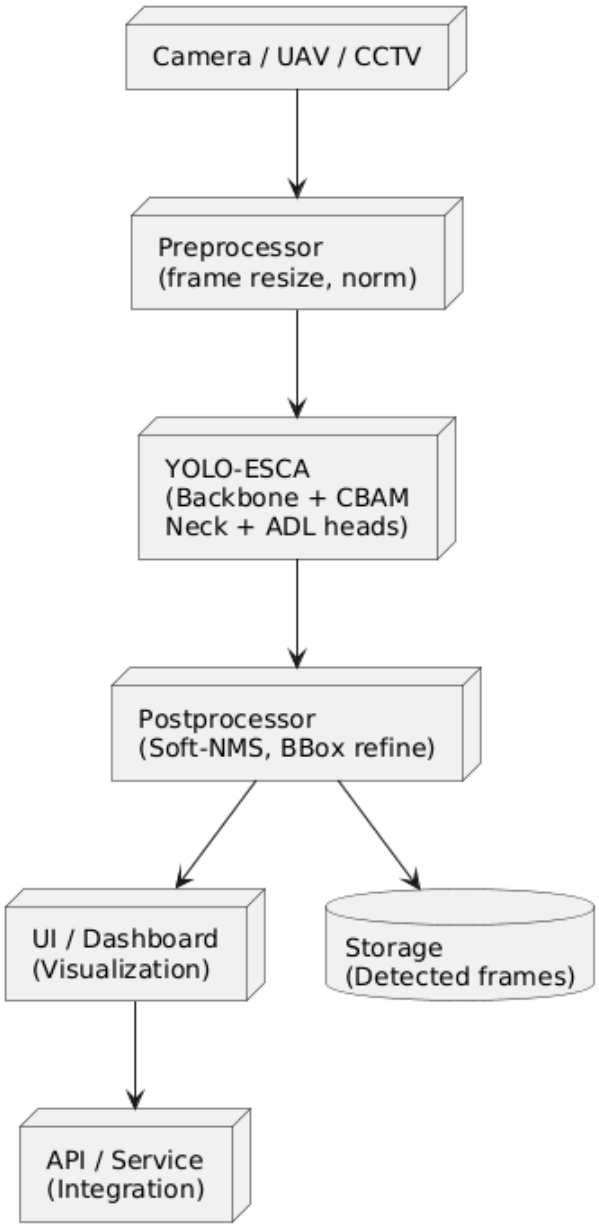


Figure 6.1.1: System Block Diagram

The system block diagram illustrates the complete pipeline of the Advanced YOLO-ESCA framework. Input video frames are pre-processed and passed into the YOLO-ESCA inference engine (backbone with CBAM, neck, detection heads including ADL). Outputs are postprocessed (Soft-NMS), visualized on UI, and violation frames are automatically stored in the storage module. The API/Service layer facilitates EHR/CMS integration and remote monitoring from UAV/edge devices.

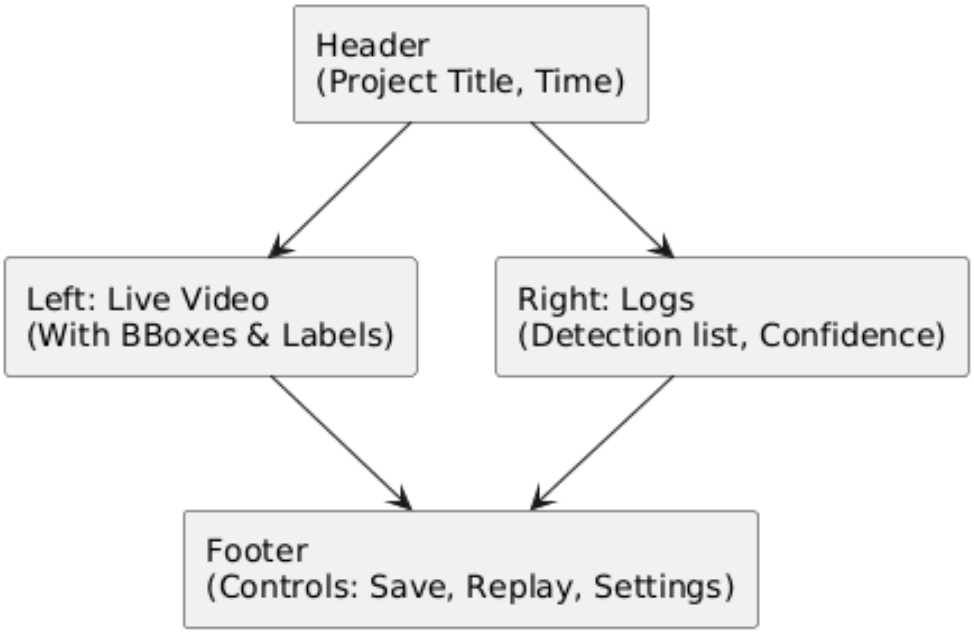


Figure 6.1.2: Wire frame diagram

This wireframe shows the expected user interface layout: live video panel with bounding boxes and class labels, a side panel with detection logs/confidence scores, and controls for saving/replaying saved violation frames..

6.2 UML Diagrams

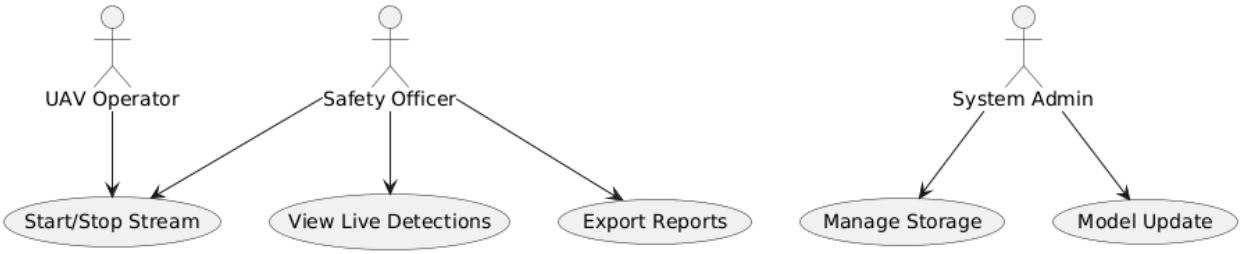


Figure 6.2.1: Use Case Diagram

This diagram highlights primary actors (Safety Officer, System Admin, UAV Operator) and their main interactions: start stream, view detections, export reports, tune model, and manage storage.

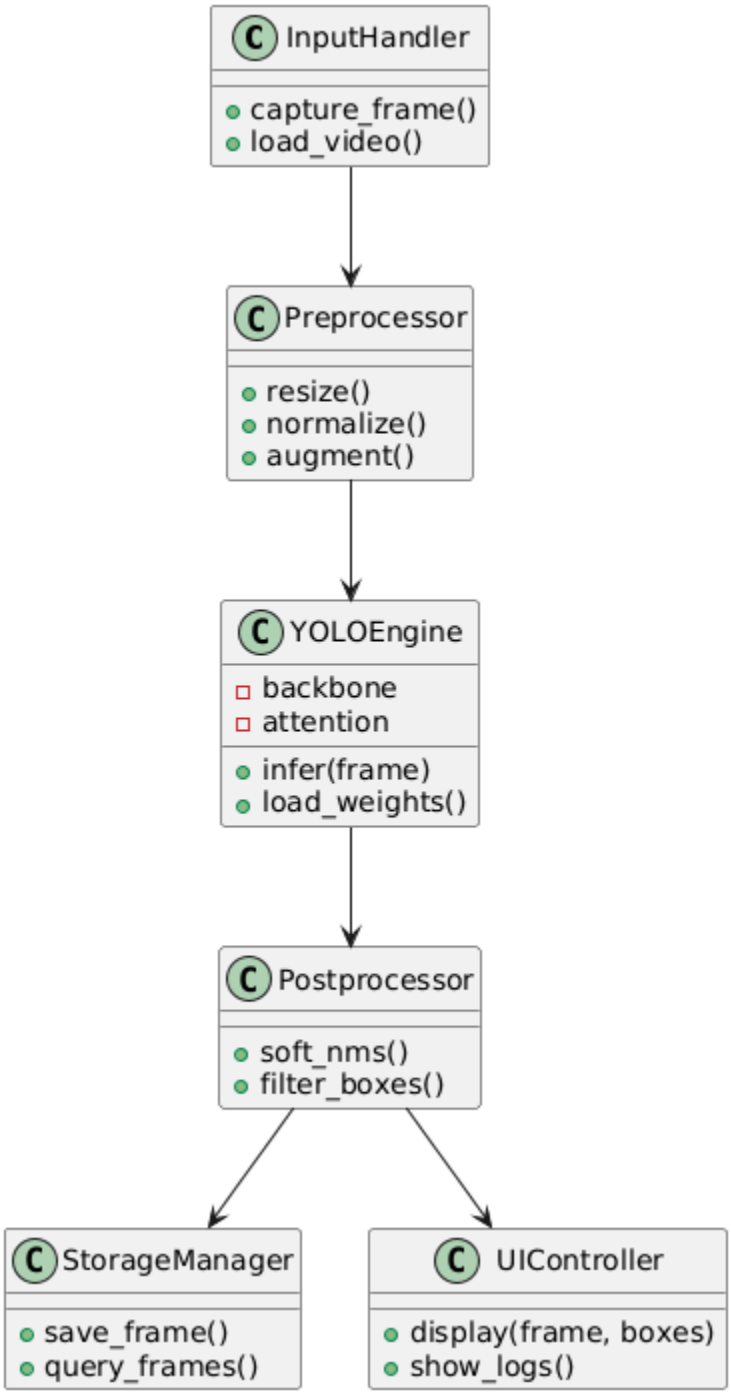


Figure 6.2.2: Class Diagram

This Class diagram describes main system classes: Input Handler, Preprocessor, YOLO Engine, Attention Module, Detector, Postprocessor, Storage Manager, API Server.

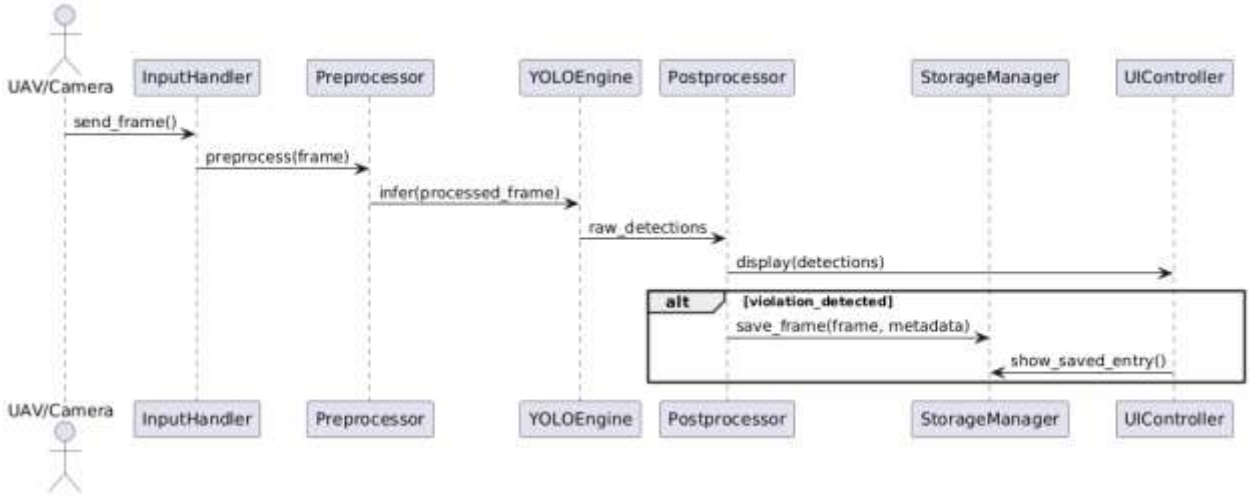


Figure 6.2.3: Sequence Diagram

The sequence diagram demonstrates the step-by-step execution of the system. A request begins when a safety officer initiates a detection process. The input frame is captured and pre-processed before being passed to the YOLO-ESCA model. The model performs prediction, applies Soft-NMS, and forwards the refined results to the output manager. If the system detects violations, the Storage Manager saves the corresponding image. Finally, results are displayed on the interface for user review

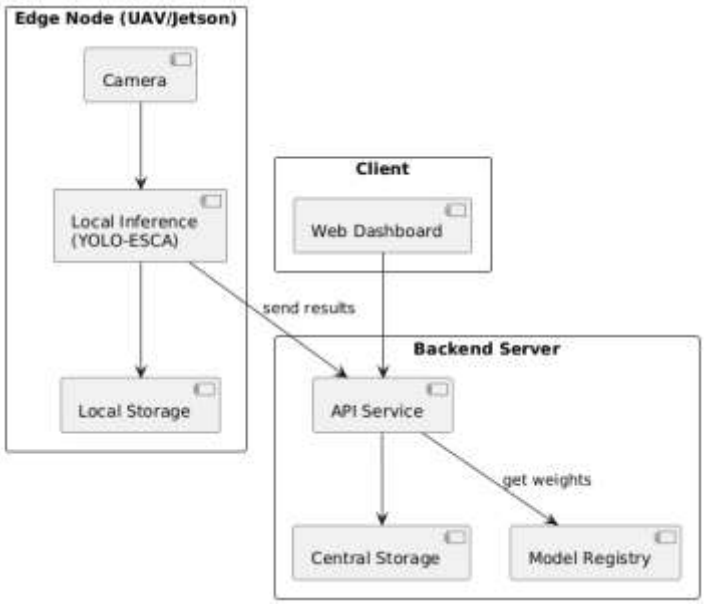


Figure 6.2.4: Component diagram

The component diagram depicts the high-level architecture of the system. It includes modules such as the Detection Engine, Attention Module, Feature Extractor, Storage Layer, Interface Layer, and Integration components for external feeds like CCTV and drones. Each component is modular and independently upgradable, ensuring system maintainability and scalability.

7. CONCLUSION

The Advanced YOLO-ESCA Framework for Enhanced Helmet Wearing Compliance Detection presented in this project aims to address the persistent challenges associated with monitoring safety-helmet usage on construction sites. Traditional manual inspection methods are inefficient, highly dependent on human attention, and prone to oversight—especially in dense working environments and long-range surveillance scenarios. The proposed framework incorporates significant architectural improvements inspired by the YOLO-ESCA model discussed in the base paper, ensuring better detection performance under complex, real-world conditions.

To enhance localization accuracy, the system adopts the Efficient Intersection over Union (EIOU-Loss), which optimizes bounding-box regression by reducing width and height discrepancies more effectively than CIOU-Loss. The inclusion of Soft-NMS helps reduce the missed-detection rate caused by target overlap in crowded scenes, ensuring that relevant bounding boxes are retained with decayed confidence rather than being removed prematurely. The integration of the Convolutional Block Attention Module (CBAM) strengthens feature extraction for small and low-visibility elements such as chin straps. Furthermore, the addition of a small-target detection layer improves the model's robustness for long-distance or drone-captured imagery.

Overall, the improved framework demonstrates strong potential for real-time deployment due to its lightweight architecture and high FPS values—consistent with the needs of UAV-based monitoring and construction-site safety systems. While limitations such as occasional missed detections in extreme low-resolution scenarios still exist, the enhancements implemented in this Stage-1 project lay a strong foundation for future work. Subsequent stages will focus on dataset expansion, model refinement, and integration into live surveillance pipelines to achieve more stable, adaptive, and context-aware safety-management solutions.

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